

SANCHARI KUNDU

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MOTIVATION

Plant single-cell biologist developing single-cell and spatial transcriptomic approaches, working at the interface of wet-lab and dry-lab, using Linux, R, and Python. Current work spans single-cell transcriptomics: annotation-refinement method for non-model plants and salt-stress responses in pennycress roots. Additional research focuses on spatial transcriptomics, developing a pathway-enrichment-based cell-type annotation in pepper. I am seeking a postdoctoral position in a computational plant genomics group to extend my current expertise toward comparative, pathway-aware annotation of crop genomes and single-cell atlases, linking gene function to stress resilience. My long-term goal is to lead an independent research group applying omics approaches to food crops and medicinal plants to support food security and human health.

RESEARCH EXPERIENCE

Virginia Tech

May 2023-present

Cell-type-specific transcriptional responses to salt stress in pennycress roots

- Generated and analyzed pennycress root snRNA-seq under salinity, resolving cell-type responses and alternative splicing/polyadenylation (AS/APA) isoform diversity
- Integrated bulk RNA-seq temporal deconvolution to build transcriptional trajectory across multiple time points
- Root imaging in wild-type and mutant pennycress using confocal and brightfield microscopy
- Whole-plant growth-chamber experiments in wild-type pennycress under controlled conditions

Refining 3' UTR annotation to improve scRNA-seq quantification in ten non-model plants

- Built a computational approach that refines 3' UTR annotation to improve transcript quantification

Cross-species spatial annotation method for pepper (Stereo-seq)

- Developing a spatial annotation approach for cell types and tissue domains using pathway-specific enrichment

Expanding a cross-species cell-type annotation resource (OMG2.0)

- Scaling Orthogroup Marker Gene (OMG) annotation method, broadening cross-species cell-identity mapping
- Developing web resource with organ- and condition-specific marker features (leaf, root, stem, stress)

Translating single-cell data into a high-resolution spatial map using fluorescent marker genes

- Confocal imaging of Arabidopsis roots with fluorescent marker lines to build a 2D root model
- Mapping of fluorescent marker-tagged Arabidopsis scRNA-seq data using Linux and R

University of New Hampshire

2018-2022

Role of putrescine and spermidine in salt and cadmium tolerance in hybrid poplar (clone NM6)

- Quantification of gas exchange by LI-COR 6400; sugars, polyamines, phytochelatin, and amino acids by HPLC

EDUCATION

PhD in Crop and Soil Environmental Sciences, SPES dept, Virginia Tech, Blacksburg, VA

2023 - present

Topic: Dissecting cell-type specific transcriptional response under abiotic stress. GPA: 3.86

MS in Integrated and Organismal Biology, DBS dept, University of New Hampshire, NH

2023

Topic: Role of polyamines in relation to nitrogen and carbon assimilation under salinity stress in poplar plants. GPA: 3.73

MSc in Agricultural Biotechnology, Ramakrishna Mission Vivekananda University, India

2016

Topic: An In-silico study of Retrotransposons in a Bamboo species (*P. edulis*), C.G.P.A - 8.14/10

BSc in Microbiology, West Bengal University of Technology, Kolkata, India

2014

Topic: Isolation of Macrophage and T-lymphocytes from *V. cholerae* infected Mice, C.G.P.A - 8.86/10

PUBLICATIONS

Book Chapter

1. **Kundu S**, Chau T.N, Maroof D, Li S (2025). Annotation of single cell clusters using marker genes within and across species. *Guide to Plant Single-Cell Technology: Functional Genomics and Crop Improvement*. <https://doi.org/10.1016/B978-0-443-23736-2.00015-0>
 - first author

Papers

2. **Kundu S**, Gagne M, Weerasinghe M, Minocha S (2026). Exogenous putrescine promotes growth under salt stress in a hybrid poplar, clone NM6. *Frontiers in Plant Science*. <https://doi.org/10.3389/fpls.2025.1641288>
 - first author
3. Haghani M, Chau TN, Alajoleen R, Chen C-Y, Dip SA, Gurung V, Hsu C-W, Kittivorawong C, **Kundu S**, Li X, Luo J, Salehin M, Simenc M, Yun J, Bargmann BOR, Li S (2026). Cross-species plant single-cell analysis: community challenges and shared solutions. *Journal of Experimental Botany*. <https://doi.org/10.1093/jxb/erag218>
4. Chau TN, Timilsena PR, Bathala S, **Kundu S**, Bargmann BOR, Li S (2025). Orthologous marker groups reveal broad cell identity conservation across plant single-cell transcriptomes. *Nature Communications*. <https://doi.org/10.1038/s41467-024-55755-0>
5. Sivaramakrishnan U, **Kundu S**, Bargmann BOR, Li S (2024). SAMPLS: A prompt engineering approach using Segment-Anything-Model for plant science research. *International Symposium on Visual Computing*. https://doi.org/10.1007/978-3-031-77389-1_14
 - co-first author
6. Miao L, Dhulipalla K, **Kundu S**, Zheng B, Li S, Kojima S (2024). Leveraging Machine Learning for Predicting Circadian Transcription in mRNAs and lncRNAs. *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. <https://doi.org/10.1109/BIBM62325.2024.10822684>

In preparation

7. FLORAAGENT: Enhancing scRNA-seq Quantification in Plants Through Refined 3' UTR Annotation Chaudhury M, **Kundu S**, Li S
 - co-first author
8. Single-nucleus transcriptomics reveal cellular remodeling during salt stress in *Thlaspi arvense* roots **Kundu S**, Gautam B, Sedbrook J, Bargmann BOR, Li S
 - first author
9. AI-enhanced data analysis in plant proteomics (Invited book chapter: *Plant Proteomics*, Elsevier) **Kundu S**, Li S
 - first author

ORAL PRESENTATIONS

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|---|------|
| 1. 3 rd Annual Discover Symposium, Salk Institute for Biological Studies, CA | 2026 |
| Cell-type specific transcriptional responses to salt stress in pennycress using single nuclei transcriptomics | |
| 2. Mid-Atlantic ASPB, University of Maryland | 2026 |
| Studying cell-type specific transcriptional responses to salt stress in pennycress roots | |
| 3. Graduate and Professional Student Senate, Virginia Tech | 2026 |
| Single-nucleus transcriptomics reveal cellular remodeling during salt stress in pennycress | |
| 4. Bayer Workshop Flash talk, Virginia Tech | 2026 |
| Cell-by-cell blueprint of salt stress adaptation in pennycress with single-nuclei transcriptomics | |
| 5. Gordon Research Conference: Single cell approaches in Plant Biology | 2025 |
| Improving scRNA-seq quantification in plants through refined 3' UTR annotation | |
| 6. IUFRO Eighth International Poplar Symposium, Serbia, Selected | 2022 |
| Effects of putrescine leaf spray in the response of hybrid poplar plants to salt stress | |
| 7. 3 Minute Thesis Competition, UNH | 2021 |
| Developing productive poplars for the harsh environment | |
| 8. Visions Board Party: "Be the Best You", OISS, UNH (Invited) | 2021 |
| A woman who inspires you – Who they are and how they inspire you | |
| 9. Department of Biological Sciences Departmental Seminar, UNH | 2020 |

POSTER PRESENTATIONS

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| 1. Mid-Atlantic ASPB, University of Maryland | 2026 |
| Single-nucleus transcriptomics reveal cellular remodeling during salt stress in pennycress | |
| 2. Translational plant Science Symposium, VT | 2026 |
| Spatiotemporal Mechanisms of Salt Stress Response in Pennycress Roots | |
| 3. Plant and Animal Genome Conference, San Diego, USA | 2026 |
| Single-nucleus transcriptomics reveal cellular remodeling during salt stress in pennycress | |
| 4. Gordon Research Conference: Single cell approaches in Plant Biology | 2025 |
| Improving scRNA-seq quantification in plants through refined 3' UTR annotation | |
| 5. School of Plant and Environmental Sciences Symposium, VT | 2025 |
| Enhancing scRNA-seq quantification in non-model plants via refined 3' UTR annotation | |
| 6. TPSC Symposium, VT | 2025 |
| Fine tuning marker gene detection in non-model plants: The power of single cell 3'UTR annotations | |
| 7. Fralin Open House, VT (Invited) | 2025 |
| Imaging plant roots in a new light: Artificial intelligence-driven segmentation with confocal microscopy | |
| 8. Plant and Animal Genome Conference | 2024 |
| Cross species single-cell annotation with orthologous marker gene groups | |
| 9. SPES Symposium, VT | 2023 |
| Cross species single-cell annotation with orthologous marker genes browser | |
| 10. American Society of Plant Biologists, Oregon | 2022 |
| Exogenous putrescine promotes growth under salt stress in a hybrid poplar | |
| 11. Frontiers in Biotechnology, St Xavier's College (Autonomous), India | 2016 |
| An in-silico study of retrotransposons in a bamboo species | |

SKILLS AND EXPERTISE

Wet Lab

- RNA extraction (pennycress, sagebrush)
- Plant tissue culture (poplar, rice, *Centella*)
- Microbiology (inoculation, serial dilution)
- Bacterial plasmid isolation
- Confocal, DIC imaging
- Hyperspectral imaging
- HPLC (sugar, polyamine, phytochelatin, amino acid)
- LI-COR 6400, LI-6800, SPAD meter
- Thermal cycler

Computational & Programming

- Linux/HPC, R (advanced), Python (proficient)
- Reproducible workflows, Git/GitHub, data visualization

Single-cell & Omics Analysis

- scRNA-seq/snRNA-seq analysis (Seurat, Scanpy); experimental design
- Spatial transcriptomics (Stereo-seq); cross-species annotation (OMG)
- Isoform & polyadenylation analysis
- 3' UTR annotation refinement; bulk RNA-seq
- Image segmentation pipelines (PlantSeg, SAMPLS, RootPainter, CellPose, Rootscope)

TEACHING

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| • <i>Virginia Tech</i> : Applied Statistics for Plant and Environmental Sciences | Spring 2025 |
| • <i>University of New Hampshire</i> : Intro to Botany, Intro to Germs, Human Anatomy and Physiology | 2018-2022 |

RELEVANT COURSEWORK

Artificial Intelligence in Agriculture, Applied Statistics for Plant and Environmental Sciences, Introduction to Genomic Data Science, Application of Advanced AI in Plant Genomics, Building Multicultural Competency in Ag & Life Sci, Scientific Skills, Mol Cell Bio, Research Ethics, Adv Study/Plant Cell & Tissue, Scientific Writing, Experimental Design & Analysis, Model organism Research, Cell Signaling Networks, Plant Genetic Engineering

HONORS & AWARDS

- Travel award to present at Mid-Atlantic ASPB 2026
- 3rd prize in Oral presentation at Graduate and Professional Student Senate Symposium 2026
- TPSC non-PI grant for research training 2026
- Travel grant from College of Agriculture, VT to present at PAG 33 2026
- 1st prize in Shark tank presentation for Introduction to AI in Agriculture 2025
- 3rd prize in Poster presentation at School of Plant and Environmental Sciences Symposium, VT 2025
- Travel grant from GPSS Travel grant program, VT to present Gordon Research Conference 2025
- Travel grant from College of Agriculture, VT to present Gordon Research Conference 2025
- Honorarium for poster presentation at Fralin Open House, VT 2025
- Funding from HHMI and Helmsley Charitable Trust for Frontiers and Techniques in Plant Science, CSHL 2024
- Travel grant award from American Society of Plant Biologists 2022
- Honorarium from Junior Sciences and Humanities Symposium 2021
- Full funding for the second year during MS program, India 2015
- 2nd position (Silver medal) in Bachelor of Science at West Bengal University of Technology, India 2014

PROFESSIONAL TRAINING

- Cold Spring Harbor Laboratory, NY* 2024
- Frontiers and Techniques in Plant Science
- Virginia Tech University, VA* 2023-2024
- Conflict of Interest, NSF- Responsible Conduct of Research Training, Graduate Honor System Panelist Training

PROFESSIONAL SERVICE

- Leadership team*
- Summer Single Cell Workshop, Virginia Tech 2024-2025
- Peer review Activities*
- Manuscript reviewer: Reviewed 2 manuscripts submitted in The Plants 2024
- Review Committee, VT*
- Reviewer: TPSC Grant Proposal, Travel Fund Program 2025
- Graduate Honor System, VT* 2024-present
- Panelist: Reviewed Academic Integrity cases as part of the GHS board
- Judging*
- Evaluated STEM research papers at Junior Sciences and Humanities Symposium 2020-2023
 - Poster evaluation at 31st COLSA Undergraduate Research Conference 2019

MENTORING

- Mentored 3 MS students (CS) for building a deep-learning image-segmentation pipeline for automated root phenotyping from confocal images (YOLO, PlantSeg)
- Supervised an undergraduate researcher on studying tomato growth response to fertilizer treatment under controlled environment conditions for hyperspectral image analysis
- Mentored 6 students in India with cover letter, SOP and CV writing for PhD applications in US universities
- Mentored high school students for Project SMART and Science Research and Internship Program (SRIP)

PROFESSIONAL MEMBERSHIP

- American Society of Plant Biologists
- PlantGene
- Phi Kappa Phi Honor Society

UNIVERSITY SERVICE AND OUTREACH

- Orientation Leader for International Students - Office of International Students and Scholars (OISS), UNH
- Graduate Student Senate, UNH
- Career Mentor at Career and Internship Fair, UNH
- Earth Day Service Buckthorn Blitz, Durham, NH

REFERENCES

Song Li, PhD, Professor

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